

Introduction/Intellectual Merit: The progression of artificial intelligence has always hinged on a delicate balance between **algorithms and evaluation**. Periods of rapid innovation have consistently followed the creation of *difficult, value-laden benchmarks*: evaluations that crystallize what society deems “intelligent.” Two notable examples are **ImageNet [5]**, which not only catalyzed an explosion of novel architectures, but also demonstrated the power of large-scale data, and **SWE-Bench [8]**, which ushered in the modern era of agentic systems by leveraging the inherently verifiable nature of code. Benchmarks like MATH [9], HumanEval [10], and Tau-Bench [11] further highlight the importance of clarity and reproducibility of evaluations for driving progress.

Yet, all of these benchmarks share a crucial limitation: they evaluate **autonomous task completion**. As a result, today’s models excel at solving **well-specified, self-contained problems**, from competitive programming to math proofs, where all relevant context is static, fully observable, or retrievable. This has driven remarkable progress in automation but has left us ill-equipped for the **next frontier of AI: collaboration with humans**. Humans are complex, stochastic, and diverse. They possess internal utility functions that are inconsistent, shifting, and often indescribable in language: a large distribution shift from the objectives our models are trained to optimize.

As AI becomes more pervasive, what society increasingly needs are not models that *replace* humans, but models that **enhance human productivity, creativity, and reasoning**. However, our evaluation paradigms, and thus our research incentives, still optimize for the former. What perpetuates this imbalance is not just conceptual, but infrastructural: autonomous benchmarks are easy to iterate on + reproduce, while human-in-the-loop evaluation is (*extremely*) expensive and difficult to scale. The result is a world in which only large corporations possess the resources to align AI with human values, often in ways that **reflect private incentives rather than public good** and with *significant lack of regulation*.

To remedy this, my research will focus on building **computational models of human behavior**: models that capture the diversity, stochasticity, and evolution of human behavior. By optimizing models **with humans in the objective**, we can achieve scalable, computational alignment for collaborative agents over long-horizon, open-ended interactions. Just as ImageNet and SWE-Bench catalyzed progress through clear, open benchmarks, human-AI research can advance through shared simulations of human behavior that are reproducible, scalable, while giving communities a direct role in shaping alignment.

Relevant Prior Work: My past research develops methods to evaluate and model human-AI collaboration to build systems that **amplify, not replace**, human agency. In *KITE (first-author, NeurIPS 2025) [2]*, we quantify knowledge transfer, measuring not just what models know, but how effectively they communicate it to humans. At Sierra, I lead the next iteration of *Tau-Bench [11]*, one of the most widely used (*1.2k github stars*) and *first* benchmarks to use **simulators of human policy** for reproducible, scalable evaluation of agent behavior. In *IMPersona (first-author, COLM 2025) [1]*, we fine-tuned language models on long-horizon conversational data to approximate individual behavioral policies.

Research Proposal: This proposal aims to build the computational, methodological, and evaluative infrastructure needed to model human behavior: also termed **user simulators**, and use these models to train and evaluate AI systems designed for *collaboration*, not automation. 3 Objectives:

Objective 1: Formalize and evaluate north-star metrics for user simulators. Before improving user simulators, we must define what makes them good. Although “user simulation” has appeared in dialogue systems and reinforcement learning, its scope remains underspecified for modeling human behavior at scale. Building on Tau-Bench’s reproducibility principles, I will develop standardized environments that compare model-predicted behaviors to human reference data across multi-turn, task-oriented settings. Metrics will capture how realistically simulators reproduce human decisions (fidelity), how well they represent heterogeneous users (diversity), and how effectively they evolve as user knowledge or preferences change (adaptivity). To operationalize these dimensions, I will model interaction as a partially observable Markov decision process (POMDP), where agents infer latent states such as beliefs, goals, and knowledge gaps, under uncertainty. This enables evaluation through *asymmetric knowledge profiles* and *latent preference modeling*, directly testing a model’s *theory-of-mind competence*: its ability to infer and act upon hidden user states. The outcome will be an open benchmark

suite defining the primitives of user simulation, establishing standardized metrics and protocols that ground future methodological and empirical advances in the field.

Objective 2: Develop methods to train more realistic and adaptive user simulators. Once evaluation standards are defined, the next goal is to develop training methods that produce user simulators capable of realistic, long-horizon adaptation. This work will explore two complementary directions: (1) **Improving trait-based simulation** (*the bulk of existing work* [11]), where models are prompted to embody described human characteristics; and (2) **Advancing policy-based simulation** (*a heavily underexplored area* [1]), where behavioral dynamics are learned directly from interactional data. Building on my prior work with IMPersona on policy-based simulation, I will explore hybrid training methods that combine these strengths. Specifically, I will investigate: (1) **Trait-conditioned policy learning**, where models are post-trained to exhibit desired traits while remaining responsive to contextual shifts. (2) **Negative sampling and counterfactual supervision** to simulate bounded rationality and knowledge gaps. (3) **Theory-of-mind fine-tuning**, teaching models to reason about latent beliefs and evolving user states. (4) **Behavioral regularization (and potentially unlearning)** to preserve diversity and prevent overfitting to narrow populations. The goal is that these approaches will produce user simulators that are both *steerable and adaptive*: models that can be directed through high-level descriptions yet still capture the richness, *inconsistency*, and dynamism of real human behavior.

Objective 3: Construct scalable, public benchmarks that evaluate human–AI collaboration. Finally, I will *integrate these user simulators* into **open, collaborative evaluation environments** that measure how well AI systems reason, adapt, and co-evolve with simulated humans. These environments move beyond static benchmarks to evaluate *collaborative competence*: how effectively a model achieves shared goals with a human partner. This builds directly on my experience leading the 3rd iteration of *Tau-Bench* [11]. I will extend this paradigm to more open-ended domains, such as creative problem-solving, tutoring, and co-writing, and leverage recent advances in reinforcement learning (RL) for language models to train agents that optimize *with humans in the objective*.

Intellectual Merit:

This proposal advances the foundations of scalable alignment by establishing a framework for developing computational models of human behavior. By enabling models to learn through interaction with open, realistic, evolving human simulators, it provides a roadmap for optimizing models with human values, stochasticity, and adaptivity in the objective. Bridging reinforcement learning, probabilistic modeling, and computational social science, it unites methodologies across AI, computational social science, and human-computer interaction to create a unified foundation for alignment through interaction.

Broader Impacts:

This proposal will democratize human-AI alignment by creating open, reproducible simulations of human behavior that make collaboration research accessible beyond large corporations. By developing computational models that capture human diversity, long-horizon reasoning, and adaptation, this work enables scalable evaluation of how AI systems assist and align with humans over time. Embedding *human values* at this multi-turn, behavioral level directly helps **mitigate long-term and existential risks** as models evolve into autonomous agents. A commitment to open benchmarks and evaluation frameworks will provide policymakers, researchers, and educators with transparent, empirically grounded tools to measure alignment, fairness, and human learning outcomes.

References: [1] Shi, Quan, et al. "IMPersona: Evaluating Individual Level LM Impersonation." *COLM* (2025). [2] Shi, Quan, et al. "When Models Know More Than They Can Explain: Quantifying Knowledge Transfer in Human-AI Collaboration." *NeurIPS* (2025). [3] Shi, Quan, et al. "Can Language Models Solve Olympiad Programming?." *COLM* (2024). [5] Russakovsky, Olga, et al. "Imagenet large scale visual recognition challenge." *IJCV* 115.3 (2015): 211-252. [8] Jimenez, Carlos E., et al. "Swe-bench: Can language models resolve real-world github issues?." *arXiv:2310.06770* (2023). [9] Hendrycks, Dan, et al. "Measuring mathematical problem solving with the math dataset." *arXiv:2103.03874* (2021). [10] Chen, Mark, et al. "Evaluating large language models trained on code." *arXiv:2107.03374* (2021). [11] Yao, Shunyu, et al. "\$\tau\$-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains." *arXiv:2406.12045* (2024)